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Scholar Augment: An End-to-End Multi-LLM Platform for Accelerating Systematic Literature Reviews and Semantic Corpus Analysis

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***Abstract*— Systematic literature reviews and large-scale qualitative document analysis are foundational to academic progress in fields ranging from Computer Science to the social sciences. However, the process is fundamentally constrained by the time-intensive, laborious, and error-prone nature of manual data extraction. This bottleneck not only slows the pace of research but also limits the scope and scale of questions that can be feasibly investigated. Existing digital tools offer partial solutions but fail to address the core challenge holistically. Standard reference managers excel at citation management but are content-agnostic, while Qualitative Data Analysis Software (QDAS) still relies on extensive manual coding.**

This paper introduces Scholar Augment, a novel, user-centric web platform designed to bridge this methodological gap. We present its architecture and a new framework for researcher-AI collaboration that accelerates systematic data extraction while keeping the researcher in full control. Scholar Augment introduces a unified pipeline that automates the entire SLR lifecycle: from AI-driven Boolean search string generation and automated PDF retrieval (via Unpaywall), to structured data extraction and semantic corpus interaction. Leveraging a provider-agnostic Multi-LLM architecture and Retrieval-Augmented Generation (RAG) backed by a vector database, the platform enables researchers to generate custom analytical frameworks. The platform incorporates a persistent database and post-processing tools for data standardization. A case study analyzing 592 academic articles demonstrates that Scholar Augment reduced data extraction time by approximately 99.32%, offering a new paradigm for efficient, scalable, and rigorous document analysis. To demonstrate real-world viability, the platform has been deployed as a public web service (https://starlit-brioche-15a023.netlify.app) and is currently actively used by an initial cohort of researchers, offering a new scalable paradigm for rigorous document analysis

Keywords-Artificial intelligence; Natural language processing; Text mining; Information retrieval; Human computer interaction; Software tools

I. Introduction

The exponential growth of academic publications presents a significant challenge for researchers across all disciplines. While this information deluge offers unprecedented opportunities for knowledge synthesis, it also creates a practical bottleneck for one of research's most crucial activities: the literature review. As Cabrera and Cabrera note in [1], a literature review is the foundational step that precedes nearly every significant scholarly output, grounding new knowledge in previous work.

In response to this challenge, many fields have championed the Systematic Literature Review (SLR) as a rigorous, scientific methodology. Lame defines SLRs as a method for synthesizing evidence to answer a specific research question in a transparent and reproducible manner [2]. Unlike traditional narrative reviews, an SLR treats the review itself as a scientific process, with explicit protocols.

However, the manual execution of an SLR remains a formidable obstacle. The process involves multiple laborious stages: developing a search strategy, screening titles and abstracts, assessing full-text articles, and extracting data. A review of published SLRs found that, on average, only 2% of identified abstracts are ultimately included, illustrating the immense manual effort required to filter relevant signal from noise [3]. The manual execution of an SLR remains a formidable obstacle. The process involves multiple laborious stages: developing complex, database-specific Boolean search strategies, manually tracking down and downloading Open Access full-text PDFs, and finally, the cognitive load of data extraction. Existing tools fragment this workflow.

Existing digital tools offer only partial solutions. Reference managers (e.g., Zotero) are "content-agnostic," while QDAS tools (e.g., NVivo) still require manual coding. General-purpose Large Language Models (LLMs) lack a systematic workflow, persistent storage, and batch processing capabilities.

This paper introduces Scholar Augment, a novel web-based platform designed to fill this gap. Scholar Augment empowers researchers by using an LLM (Google's Gemini) to automate structured data extraction. Its core innovation is a workflow allowing researchers to define schemas using natural language, apply them to document batches, and curate results via post-processing tools.

The primary contribution of this paper is the introduction of Scholar Augment, a novel software application designed specifically to help the academic community accelerate systematic literature reviews. Rather than proposing a theoretical training methodology, we present a fully functional

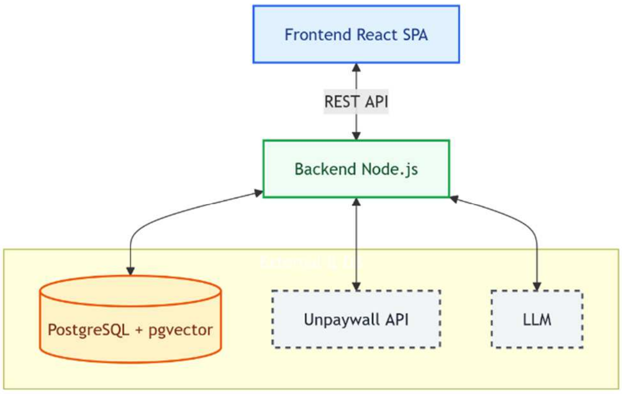

Fig. 1 High-level system architecture of Scholar Augment, illustrating the separation between Client, Server, and External Data layers.

tool and detail its built-in data analysis workflow. To validate the application's real-world utility, we present a case study analyzing 592 articles on wearable technology, followed by a discussion of the system's architecture and performance.

II. The Scholar Augment Platform

A. *Researcher in Control*

The fundamental principle of Scholar Augment is to augment, not replace, the researcher. While the platform automates repetitive tasks, the intellectual authority remains with the user. The researcher acts as the architect of the classification schema, the curator of the extracted data, and the sole interpreter of the results. Every feature operates under explicit user control to ensure academic rigor.

B. *System Architecture*

To ensure accessibility and scalability, Scholar Augment is implemented as a modern full-stack web application. The architecture follows a client-server model, decoupling the user interface from business logic and data persistence, as illustrated in Fig. 1.

The system is composed of three primary, interacting components. The Frontend Client is a dynamic single-page application (SPA) built with the React library. It is responsible for rendering the entire user interface, managing local UI state, and communicating with the backend via secure API requests. This client-side application encompasses all user-facing elements, including the dashboard, schema creation forms, the classification interface, and the results visualization pages.

The Backend Server functions as the application's core, consisting of a secure RESTful API built with Node.js and the Express framework. It handles all business logic, including user authentication via JSON Web Tokens (JWT), processing API routes, and managing all database transactions through the Sequelize Object-Relational Mapper (ORM). A key function of the backend is to act as a secure intermediary for all communications with external services, ensuring that sensitive credentials like API keys are never exposed to the client. The Database and External Services layer handles data persistence, vector embeddings, and third-party integrations. The platform employs a PostgreSQL relational database extended with pgvector to store both structured application data and high-dimensional semantic embeddings (Document Chunks) for advanced text retrieval. Unlike systems locked into a single AI provider, Scholar Augment implements a dynamic, provider-agnostic Multi-LLM adapter, securely routing user requests to Google Gemini, OpenAI, Anthropic, or DeepSeek based on the specific cognitive demands of the task. Additionally, it integrates the Unpaywall REST API for automated open-access article retrieval.

C. *Features and Workflows*

To address the fragmentation in current digital tools, Scholar Augment standardizes the literature review process into a cohesive, four-step application workflow (as depicted in Fig. 2):

- **Step 1:** The workflow begins with the Search String Builder. The user describes their research question, and the LLM acts as an expert librarian, generating optimized Boolean syntax for multiple databases. Once DOIs are identified, the platform’s DOI Fetcher automatically locates and downloads legally available Open Access PDFs directly to the user's local environment via the Unpaywall API.

- **Step 2:** The platform removes the friction of starting a new analysis by allowing users to define a custom classification schema using natural language. The system's LLM generates a structured system prompt containing the desired data categories, which the user saves as a reusable analytical framework.

- **Step 3:** The user uploads their corpus of PDFs. To optimize performance, the application first performs a SHA-256 contentHash check to avoid re-processing identical documents. The extracted text is then sent to the multi-LLM adapter alongside the custom schema, and the structured JSON output is saved to the PostgreSQL database.

- **Step 4:** Recognizing that raw AI output can contain semantic variances, the final step involves a post-processing interface. The researcher maps synonymous terms (e.g., merging "Audio" and "Auditory" into a single canonical category). This transforms raw AI extraction into a clean, standardized dataset ready for quantitative analysis and export. Additionally, users can interact with their customized corpus using the RAG-powered "Chat with Corpus" module for semantic querying.

D. Deployment and Early Adoption

Unlike many theoretical frameworks proposed in literature, Scholar Augment has been developed as a fully functional, production-ready SaaS (Software as a Service). The platform is publicly accessible at https://starlit-brioche-15a023.netlify.app. At the time of writing, the platform has successfully onboarded an initial cohort of 5 active researchers who are utilizing the system for their independent systematic reviews. This early adoption validates the platform's user-centric design, stability, and the immediate need for such augmentation tools within the academic community.

III. Case Study: Systematic Literary Review of the Wearable Technology for Hearing Impairment

A. Objective

To validate the Scholar Augment platform, a large-scale case study was conducted not as a simulation, but as the primary methodology for performing a comprehensive systematic literature review integral to our ongoing PhD research. The objective was to utilize the platform's full capabilities to analyze a substantial corpus of academic literature within the domain of Computer Science and Information Technology. This case study serves a dual purpose: it demonstrates the end-to-end workflow of the application while also presenting the preliminary findings of the literature review itself, made possible by the efficiency of the tool.

B. Methods

A comprehensive search was performed across three major academic databases: IEEE Xplore, the ACM Digital Library, and Web of Science. A detailed Boolean search string was developed to identify literature at the intersection of hearing disabilities, assistive technologies, and wearable devices:

(("auditory disabilit" OR "hard of hear" OR heari* (loss OR impair* OR difficult OR challenge OR issue OR deficit OR decline OR reduction OR problem OR disorder) OR deaf*) AND ("assist" NEAR/3 (technology OR device OR system OR equip OR solution OR application OR gadget OR mechanism) OR "heari* aid" OR implant) AND (wear* (device OR gadget OR tool* OR equip* OR apparat OR sys* OR accessor OR produc) OR smart OR "body-worn"))*

This structured search yielded a final corpus of **592** academic articles, which formed the basis for this systematic review.

Using the Scholar Augment platform, a custom schema titled "Wearable Tech for Hearing Loss Analysis" was created to meet the specific objectives of our research. The schema was designed to extract sixteen distinct data categories critical for understanding the current state of the field. These categories included Type of Hearing Disability, Type of Wearable Device, User Study, Number of Users, Sensory Modalities Used, and Reported Limitations/Challenges, among others, creating a detailed analytical framework for the subsequent data extraction.

C. Procedure and Results

The corpus of 592 PDF articles was uploaded to Scholar Augment, and the "Wearable Tech for Hearing Loss Analysis" schema was applied to initiate the batch classification process.

The entire end-to-end data extraction process from uploading the 592 documents to completing the AI classification for the entire corpus was completed in approximately **100 minutes**. A conservative estimate for a traditional manual data extraction process, assuming an average of 30 minutes per paper to read and record the sixteen data categories, is 17,760 minutes (296 hours).

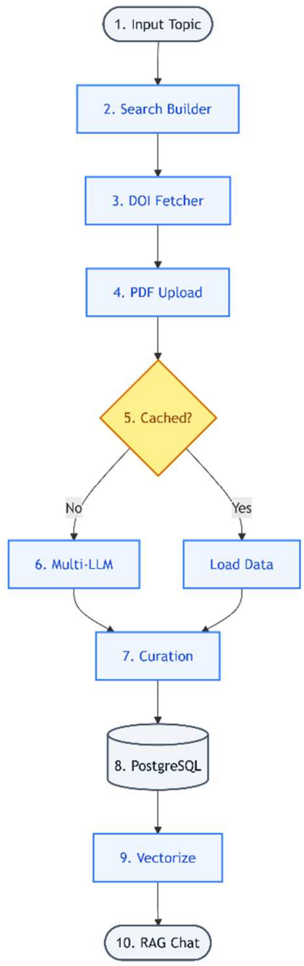

Fig.2 The logical workflow of the platform, detailing the end-to-end pipeline from automated literature retrieval and LLM-driven data extraction to semantic corpus interaction via Retrieval-Augmented Generation (RAG).

While the initial corpus for this case study was gathered manually, the subsequent integration of the Search Builder and DOI Fetcher into Scholar Augment now allows this preliminary literature gathering phase to be accelerated with the same platform.

This represents a time savings of 99.32%. This dramatic acceleration was the enabling factor that time savings of approximately made a review of this scale feasible, transforming a task that would have taken months of dedicated manual labor into a computationally-driven process completed in under two hours.

Upon completion, the raw AI-generated data was curated using the platform's integrated Standardization tool. This post-processing step was essential for ensuring data consistency. For instance, in the Age Group category, the terms "Infants," "Children," and "Young children" were all programmatically

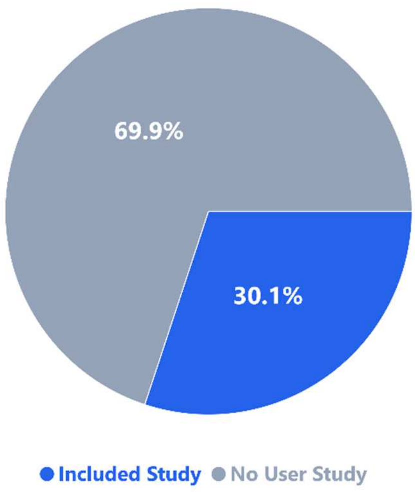

Fig. 3a. Distribution of academic articles within the analyzed corpus (N=592) based on the inclusion of primary user studies.

mapped to the single canonical category "Children". Similarly, in the Type of Wearable Device category, ten different variations such as "Wearables" and "Smart wearable device" were consolidated into the standardized term "Wearable devices". This process ensured the integrity of the dataset for subsequent analysis. The final, cleaned data was presented in the History page table, a sample of which is shown in Table. 1

The standardized dataset was then used to generate summary statistics and visualizations. For example, the distribution of papers that included a user study versus those that did not was clearly visualized (Fig 3a). The analysis revealed that a significant majority of the literature did not involve direct user studies. Furthermore, an analysis of the "Wearable Device Body Location" category showed a strong prevalence of ear-based devices, with a smaller distribution across other locations like the wrist and head (Fig 3b). These visualizations provide immediate, high-level insights into the state of the research field, which would be difficult to discern from reading the papers individually.

IV. Discussion

The results of the case study demonstrate that Scholar Augment is an effective tool for accelerating systematic document analysis. However, the implications of such a platform extend beyond mere efficiency gains. This section discusses the broader impact on research methodology, the importance of maintaining human oversight in an era of generative AI, and the future trajectory of this work.

A. From Data Extractor to Data Curator

The integration of AI into the research workflow is fundamentally redefining the role of the academic researcher. While initial concerns suggested AI might replace researchers, the consensus is shifting towards a partnership model. As highlighted by Raisch and Krakowski, the relationship between automation and augmentation presents a paradox: over-reliance on pure automation can lead to deskilling, whereas augmentation preserves human oversight while enhancing capacity [9]. Scholar Augment is strictly designed within this augmentation paradigm.

By automating the mechanical task of structured data extraction, the researcher’s effort is shifted from low-level data entry to high-level data curation. This concept is supported by Hong et al., who argue that "cognitive offloading" via generative AI allows learners and researchers to engage more deeply in higher-order thinking, such as analysis and synthesis, rather than getting bogged down in routine processing [10]. Consequently, the user of Scholar Augment evolves from a passive extractor into an active architect of the analytical framework.

B. The Necessity of "Human-in-the-Loop" (HITL)

A central principle in the design of Scholar Augment is the "human-in-the-loop" (HITL) workflow. This design choice is not merely a feature but a necessity for academic rigor, given the

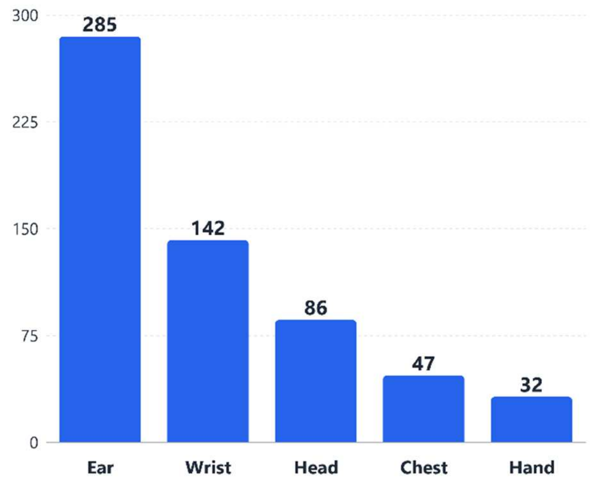

Fig. 3.b. Frequency distribution of targeted wearable device body locations across the extracted literature. ***(Note: Some articles may propose multi-location devices, resulting in overlapping counts).***

known limitations of Large Language Models (LLMs). Recent findings by Chelli et al. are alarming; they demonstrated that LLMs like ChatGPT and Bard can hallucinate references and data at rates exceeding 90% in some configurations, creating convincing but fictitious outputs [11].

Furthermore, Pitre et al. identified a phenomenon known as 'sycophancy' in AI interactions, where models tend to reinforce errors or agree with the user/other agents rather than critically correcting facts [18]. Uncritical adoption of such outputs would compromise research integrity.

C. *Comparison with State of the Art*

The landscape of text classification has evolved from traditional models (like Naïve Bayes or SVM) to deep learning approaches [15], and finally to the current generative era. While automated data extraction has been a goal for over a decade—as reviewed by Jonnalagadda et al., who noted the lack of unified frameworks [16] recent solutions still face challenges.

Scherbakov et al. conducted a systematic review using LLMs and found them promising for specific phases [8]. However, commercial tools often operate as "black boxes" with rigid schemas. In contrast, Azevedo et al. highlight in their review of Automated Machine Learning (AutoML) tools that while they democratize access to AI, they often lack flexibility for complex, domain-specific scenarios [14].

Furthermore, many commercial AI tools suffer from 'vendor lock-in' and operate as black boxes. Scholar Augment's architecture mitigates this by allowing users to 'bring their own key' (BYOK) across multiple state-of-the-art providers (OpenAI, Anthropic, Google). Additionally, while most systematic review tools focus only on metadata screening (e.g., Rayyan), Scholar Augment unites the entire lifecycle from query building and automated full-text fetching to advanced semantic querying via RAG into a single, cohesive academic suite

Scholar Augment distinguishes itself through User-Defined Schemas. Unlike tools with fixed extraction fields, our platform leverages the zero-shot capabilities of Gemini to allow researchers to define any analytical framework using natural language. Additionally, by implementing a content-hash caching mechanism (SHA-256), we address the computational cost and environmental concerns associated with redundant AI processing, offering a transparent and persistent audit trail.

Future development will focus on transforming Scholar Augment from a single-user tool into a Collaborative Research Environment. As foundational work by Churchill and Snowdon on collaborative virtual environments suggests, shared context and awareness are critical for cooperative work [17]. We aim to implement team-based schema definitions and multi-user curation pipelines to support interdisciplinary research groups.

V. Conclusion And Future Work

This paper presented Scholar Augment, a novel multi-LLM web platform designed to accelerate systematic literature reviews through researcher-AI collaboration. The platform addresses a critical bottleneck in academic research by unifying the entire SLR lifecycle into a single, cohesive workflow: from AI-driven Boolean search string generation and automated PDF retrieval, to structured data extraction via a provider-agnostic Multi-LLM adapter, semantic corpus querying through RAG, and post-processing standardization tools.

The case study conducted on a corpus of 592 academic articles on wearable technology for hearing impairment validated the platform's real-world utility, demonstrating a time reduction of approximately 99.32% compared to traditional manual extraction methods. Critically, this efficiency was achieved without sacrificing research rigor, as the human-in-the-loop design ensures that the researcher retains full intellectual authority over the analytical process. Scholar Augment thus represents a concrete realization of the augmentation paradigm described by Raisch and Krakowski [9], shifting the researcher's role from mechanical data extractor to high-level data curator and analytical architect.

The platform's current deployment as a publicly accessible SaaS application, actively used by an initial cohort of researchers, further confirms its practical viability and immediate relevance to the academic community.

Despite these promising results, several limitations must be acknowledged. The accuracy of AI-generated extractions is inherently dependent on the quality of the underlying LLM and the clarity of the user-defined schema. As demonstrated by Chelli et al. [11], hallucination remains a non-trivial risk in LLM-assisted workflows, reinforcing the necessity of the platform's human curation step. Additionally, the current system relies on user-supplied API keys, which may present a barrier for researchers unfamiliar with AI service providers.

Future development will pursue several directions. First, Scholar Augment will be extended into a collaborative research environment, allowing teams to define shared schemas, assign curation tasks, and maintain synchronized corpora drawing on the principles of collaborative virtual environments outlined by Churchill and Snowdon [17]. Second, the platform will incorporate automated quality assurance mechanisms, including cross-validation between multiple LLM providers on the same document, to detect and flag potentially hallucinated extractions before they reach the researcher. Third, we plan to expand the RAG module to support multi-document reasoning, enabling researchers to pose complex, comparative questions across the entire corpus rather than querying individual documents. Finally, integration with additional academic databases and citation management tools such as Zotero is planned, further streamlining the end-to-end SLR pipeline and reducing the remaining manual steps in the workflow.

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